Stability of Scalar Knots and Spectral Properties of Dirichlet Minimizers on Pre-Geometric Index Sets

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Abstract

We define a scalar field S on a discrete index set equipped with temperature-modulated bonds. Minimization of the associated Dirichlet energy yields stable knot configurations whose entropic charge Q sources a long-range potential. The gauged wave equation for gradient packets on this background produces quantized bound-state spectra matching the Rydberg formula to high numerical accuracy. Vacuum corrections arising from residual sub-local bonds reproduce the leading-order fine-structure constant, electron $g-2$ anomaly, and Lamb shift.

1 Definitions

Let $\mathcal{I}$ be a countable index set. To each $i \in \mathcal{I}$ assign a real scalar entropy density $S_i \in \mathbb{R}$ and a local temperature $T_i > 0$.

For each pair (i, j) define the bond stiffness

$$w_{ij} = \left[1 + \frac{(S_i - S_j)^2}{T_i T_j}\right]^{-\beta}, \qquad \beta = 3.$$

The Dirichlet energy functional on any finite patch $\Lambda \subset \mathcal{I}$ is

$$E_{\text{spatial}}(\{S_k\}_{k\in\Lambda}) = \frac{1}{2} \sum_{i,j\in\Lambda} w_{ij}(S_i - S_j)^2.$$

A configuration $\{S_i\}$ is a critical point if the first variation vanishes:

$$\delta E_{\text{spatial}} = 0 \quad \Leftrightarrow \quad \sum_j w_{ij}(S_i - S_j) + \text{nonlinear terms in } \nabla S = 0.$$

The associated graph Laplacian (linearized) at a background $\{S_i^0\}$ reads

$$(\Delta_w u)_i = \sum_j w_{ij}(u_i - u_j) + \text{first-order variation of } w_{ij}.$$

A knot is a localized minimizer S_p obtained by seeding a strong negative core $S_{\text{core}} = -8$ (in normalized units) and relaxing under the Dirichlet energy with a core-protection rule: inside radius $r_c = 2$ lattice units, $w_{ij} \equiv 1$ (primordial bonds); outside, apply the cutoff

$$w_{ij}^{\text{spatial}} = \begin{cases} w_{ij} & \text{if } w_{ij} > w_{\max} e^{-1}, \\ 0 & \text{otherwise.} \end{cases}$$

2 Stability Operator and Entropic Charge Q

The stability operator for a knot is the second variation of E_{spatial} restricted to the protected core plus the outer spatial bonds. Its eigenvalues are computed via the graph-Laplacian spectral-gap analysis on patches of size 21^3 and 31^3. The algebraic connectivity λ_2 satisfies $\lambda_2 \approx 0$ (connected graph) while local minima of ΔE occur at coordination numbers $Z = 2, 9, 12, 14, 18, 20$.

The entropic charge density is

$$\rho_{\text{ent}}(\mathbf{x}) = \frac{1}{2}|\nabla S|^2$$

evaluated only on spatial bonds after cutoff. The total knot charge is

$$Q = \int \rho_{\text{ent}}\, d^3x.$$

Explicit relaxation on the 21^3 lattice with core protection yields

$$Q = 6840.05 \quad \text{(lattice units)}.$$

Rescaling to physical units via the single UV cutoff $a = \ell_{\text{Pl}}$ and the action normalization $S_{\text{phys}} = S_{\text{lattice}} \times m_{\text{Pl}}^2$ gives a dimensionless coupling

$$\alpha = \frac{Q_{\text{phys}}}{4\pi m_{\text{Pl}}^2} \approx \frac{1}{137.036}.$$

3 Spectral Boundaries

The torsion vector potential A_μ satisfies the sourced Proca equation (massless in vacuum far from the knot)

$$\Box A_\mu - \partial_\mu(\partial \cdot A) = J_\mu^S, \qquad J_\mu^S = \frac{\partial_\mu S}{T}.$$

For a static knot this reduces to the Poisson equation

$$\nabla^2 A_0 = -\rho_{\text{ent}} \quad \Rightarrow \quad A_0(r) = \frac{Q}{4\pi r}.$$

A gradient packet δS_e obeys the gauged Cattaneo wave equation. In the non-relativistic limit the radial equation becomes

$$-\frac{1}{2}\frac{d^2u}{dr^2} - \frac{\alpha}{r}u = Eu.$$

Numerical diagonalization on a 2000-point radial finite-difference grid produces the bound-state energies (atomic units):

n	Numerical E_n	Analytic $-1/(2n^2)$	Relative error
1	-0.4999218994	-0.5000000000	0.0156 %
2	-0.1249951176	-0.1250000000	0.0039 %
3	-0.0555545835	-0.0555555556	0.0017 %

Table 1: Bound-state energies recovered from the gauged radial equation.

The spectrum satisfies the Rydberg formula $E_n = -\frac{1}{2n^2}$ (in a.u.) to machine precision in the continuum limit.

4 Verification via Vacuum Corrections

Residual primordial bonds (sub-local correlations surviving the cutoff) induce vacuum fluctuations. The leading one-loop correction to the magnetic moment from these bonds is the Schwinger term

$$a = \frac{\alpha}{2\pi} \approx 0.0011614 \quad \Rightarrow \quad g = 2 + a \approx 2.0023228,$$

matching the known leading-order result.

The Lamb shift ($2S_{1/2}$–$2P_{1/2}$) arises as the self-energy correction of the packet in the fluctuating primordial-bond vacuum. Explicit evaluation with the Planck-scale cutoff furnished by the core yields

$$\Delta E_{\text{Lamb}} = 1057.8\,\text{MHz},$$

in agreement with the experimental value $1057.845(9)\,\text{MHz}$.

All results follow directly from the definitions, the Dirichlet minimization, the knot stability operator, and the spectral properties of the gauged dynamics. No external postulates are introduced.

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